SMAPhysics.com
JEE MAIN PHYSICS DPP: HORIZONTAL PROJECTILE MOTION

Topic: Horizontal Projection: Time of Flight, Range, Velocity & Trajectory | **Time: 45 Min** | **Max Marks: 80**

Marking Scheme: **+4, -1** | Designed & Curated by: **Syed Mazhar Ali**

Q1. A projectile is launched horizontally from a height h with an initial velocity u. The time of flight T depends on:

(a) Both u and h
(b) Height h only
(c) Velocity u only
(d) Mass of the projectile

Q2. An aeroplane flying horizontally at a height of 490 m with a speed of 180 km/h drops a food packet. The horizontal distance covered by the packet before hitting the ground is: (Take $g = 9.8$ m/s^2)

(a) 250 m
(b) 1000 m
(c) 500 m
(d) 750 m

Q3. A particle is projected horizontally from the top of a cliff of height H with speed u. The equation of its trajectory taking the launch point as the origin is:

(a) $y = \frac{u^2}{2g}x^2$
(b) $y = -\frac{u^2}{2g}x$
(c) $y = \frac{g}{2u}x^2$
(d) $y = -\frac{g}{2u^2}x^2$

Q4. A ball is projected horizontally from a tower with speed $u = 20$ m/s. The angle made by the velocity vector with the horizontal after 2 seconds is: (Take $g = 10$ m/s^2)

(a) 45°
(b) 30°
(c) 60°
(d) 90°

Q5. Two bodies A and B are projected horizontally from the top of a tower with velocities u_1 and u_2 ($u_1 > u_2$) respectively. If T_1 and T_2 are their respective times of flight, then:

(a) $T_1 > T_2$
(b) $T_1 = T_2$
(c) $T_1 < T_2$
(d) $T_1/T_2 = u_1/u_2$

Q6. A stone is projected horizontally with velocity u from the top of a tower. The radius of curvature of its trajectory at the point of projection is:

(a) $\frac{u^2}{2g}$
(b) $\frac{2u^2}{g}$
(c) $\frac{u^2}{g}$
(d) $\frac{u^2}{4g}$

Q7. A ball is thrown horizontally from the top of a tower of height 80 m with speed $u = 30$ m/s. The speed with which it strikes the ground is: (Take $g = 10$ m/s^2)

(a) 30 m/s
(b) 40 m/s
(c) 70 m/s
(d) 50 m/s

Q8. A hunter aims his rifle horizontally directly at a monkey sitting on a tree branch at height h. The monkey drops from the branch at the exact instant the bullet leaves the muzzle. The bullet will:

(a) Always hit the monkey regardless of the muzzle speed u (provided u is large enough to bridge the distance)
(b) Pass above the monkey
(c) Pass below the monkey
(d) Hit only if $u = \sqrt{gh}$

Q9. A ball is thrown horizontally from the top of an inclined plane of inclination θ with speed u. The time of flight of the ball before hitting the incline is:

(a) $\frac{u\tan\theta}{g}$
(b) $\frac{2u\tan\theta}{g}$
(c) $\frac{2u\sin\theta}{g}$
(d) $\frac{2u\cos\theta}{g}$

Q10. In Q9, the distance along the incline from the launch point to where the ball lands is:

(a) $\frac{2u^2\tan\theta}{g}$

(b) $\dfrac{2u^2 \sin\theta}{g\cos^2\theta}$

(c) $\dfrac{2u^2 \tan\theta \sec\theta}{g}$

(d) $\dfrac{u^2 \tan^2\theta}{g}$

Q11. A projectile fired horizontally from a height h lands at a horizontal distance equal to $2h$. The launch speed u is:

(a) $\sqrt{gh}$

(b) $\sqrt{\dfrac{gh}{2}}$

(c) $2\sqrt{gh}$

(d) $\sqrt{2gh}$

Q12. A particle is projected horizontally from a height with speed u. The instant at which the tangential acceleration equals the centripetal (normal) acceleration is:

(a) $t = \dfrac{u}{g}$

(b) $t = \dfrac{2u}{g}$

(c) $t = \dfrac{u}{2g}$

(d) $t = \dfrac{\sqrt{2}u}{g}$

Q13. Two stones are projected horizontally in opposite directions from the top of a tower with velocities u_1 and u_2. The time after which their velocity vectors become mutually perpendicular is:

(a) $\dfrac{u_1u_2}{g}$

(b) $\dfrac{\sqrt{u_1u_2}}{g}$

(c) $\dfrac{\sqrt{u_1^2+u_2^2}}{g}$

(d) $\dfrac{u_1+u_2}{2g}$

Q14. In Q13, the separation between the two stones at the instant their velocity vectors are mutually perpendicular is:

(a) $\dfrac{u_1u_2}{g}$

(b) $\dfrac{u_1+u_2}{g}$

(c) $\dfrac{(u_1+u_2)\sqrt{u_1u_2}}{g}$

(d) $\dfrac{u_1^2+u_2^2}{g}$

Q15. A ball rolls off the top of a staircase with horizontal velocity u. If the steps are of uniform height h and width b, the ball will hit the nth step if n satisfies:

(a) $n = \dfrac{u^2h}{gb^2}$

(b) $n = \dfrac{2ub}{gh^2}$

(c) $n = \dfrac{u^2b^2}{2gh}$

(d) $n = \dfrac{2u^2h}{gb^2}$

Q16. A bomber flies horizontally at speed u at height H. A target is stationary on the ground. The angle ϕ with the horizontal at which the target must be seen by the sight-line at the moment of bomb release is:

(a) $\tan^{-1}\left(\dfrac{\sqrt{gH}}{\sqrt{2}u}\right)$

(b) $\tan^{-1}\left(\dfrac{\sqrt{2}u}{\sqrt{gH}}\right)$

(c) $\tan^{-1}\left(\dfrac{u}{\sqrt{2gH}}\right)$

(d) $\tan^{-1}\left(\dfrac{2u}{\sqrt{gH}}\right)$

Q17. A particle is projected horizontally from a height h. The displacement of the particle when its velocity vector makes an angle of 45° with the horizontal has magnitude:

(a) $\dfrac{u^2}{g}$

(b) $\dfrac{\sqrt{5}u^2}{2g}$

(c) $\dfrac{\sqrt{3}u^2}{g}$

(d) $\dfrac{2u^2}{g}$

Q18. From the top of a tower of height H, a particle is projected horizontally with speed u. The rate at which the distance r from the projection point increases initially ($t \to 0$) is:

(a) 0

(b) g

(c) u

(d) ∞

Q19. A stone is projected horizontally with velocity $u = 10$ m/s from a tower. At time $t = 1$ s, the radius of curvature of the trajectory is: (Take $g = 10$ m/s^2)

(a) 10 m

(b) $10\sqrt{2}$ m

(c) 40 m

(d) $20\sqrt{2}$ m

Q20. A particle is projected horizontally from the top of a tower. During its motion, the horizontal component of acceleration a_x and vertical component a_y are respectively:

(a) $0,\ g$ (downwards)

(b) $g,\ g$ (downwards)

(c) $0,\ 0$

(d) $g\cos\theta,\ g\sin\theta$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

SMAPhysics.com | Curated by **Syed Mazhar Ali**

Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	**6**	(c)	**11**	(d)	**16**	(a)
2	(c)	**7**	(d)	**12**	(a)	**17**	(b)
3	(d)	**8**	(a)	**13**	(b)	**18**	(c)
4	(a)	**9**	(b)	**14**	(c)	**19**	(d)
5	(b)	**10**	(c)	**15**	(d)	**20**	(a)

Detailed Solutions

Sol 1. Correct Option: (b)
Vertical motion is under gravity from rest ($u_y = 0$):

$$h = \frac{1}{2}gT^2 \implies T = \sqrt{\frac{2h}{g}}$$

T depends solely on vertical height h and acceleration due to gravity g, entirely independent of horizontal velocity u.

Sol 2. Correct Option: (c)

- $u = 180$ km/h $= 180 \times \frac{5}{18} = 50$ m/s.
- Time to fall 490 m:

$$T = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 490}{9.8}} = \sqrt{100} = 10 \text{ s}$$

Horizontal distance:

$$R = u \times T = 50 \times 10 = 500 \text{ m}$$

Sol 3. Correct Option: (d)
Taking launch point as origin $(0, 0)$:
$x = ut \implies t = \frac{x}{u}$.
$y = -\frac{1}{2}gt^2 = -\frac{1}{2}g\left(\frac{x}{u}\right)^2 = -\frac{g}{2u^2}x^2$.

Sol 4. Correct Option: (a)
At $t = 2$ s:
$v_x = u = 20$ m/s.
$v_y = gt = 10 \times 2 = 20$ m/s.
Angle with horizontal:

$$\tan\theta = \frac{v_y}{v_x} = \frac{20}{20} = 1 \implies \theta = 45^\circ$$

Sol 5. Correct Option: (b)
Since vertical heights are identical, both bodies fall under identical vertical acceleration from rest:

$$T_1 = T_2 = \sqrt{\frac{2h}{g}}$$

Sol 6. Correct Option: (c)
At the launch point: $v = u$, and acceleration perpendicular to velocity is $a_n = g$.
Radius of curvature:

$$R = \frac{v^2}{a_n} = \frac{u^2}{g}$$

Sol 7. Correct Option: (d)

- Horizontal velocity: $v_x = u = 30$ m/s.
- Vertical velocity on striking ground:

$$v_y = \sqrt{2gH} = \sqrt{2 \times 10 \times 80} = \sqrt{1600} = 40 \text{ m/s}$$

Total speed:

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{30^2 + 40^2} = 50 \text{ m/s}$$

Sol 8. Correct Option: (a)
In the vertical plane, both the bullet and the monkey experience the same gravitational downward acceleration g from the instant of release. Thus, their vertical drops in time t are identical ($\Delta y = \frac{1}{2}gt^2$), guaranteeing collision.

Sol 9. Correct Option: (b)
Let incline make angle θ with horizontal.
$x = ut$, $y = \frac{1}{2}gt^2$.
On the incline: $\tan\theta = \frac{y}{x} = \frac{\frac{1}{2}gt^2}{ut} = \frac{gt}{2u} \implies t = \frac{2u\tan\theta}{g}$.

Sol 10. Correct Option: (c)
$x = ut = \frac{2u^2\tan\theta}{g}$.
Distance along incline:

$$L = \frac{2u^2\tan\theta\sec\theta}{g}$$

Sol 11. Correct Option: (d)
$R = u\sqrt{\frac{2h}{g}} = 2h$.
Squaring both sides:

$$u^2\left(\frac{2h}{g}\right) = 4h^2 \implies u^2 = 2gh$$

$$u = \sqrt{2gh}$$

Sol 12. Correct Option: (a)
Velocity vector: $\vec{v} = u\hat{i} - gt\hat{j}$.
Acceleration vector: $\vec{a} = -g\hat{j}$.
Tangential acceleration:

$$a_t = \frac{\vec{a} \cdot \vec{v}}{v} = \frac{g^2 t}{\sqrt{u^2 + g^2 t^2}}$$

Normal acceleration:

$$a_n = \frac{gu}{\sqrt{u^2 + g^2 t^2}}$$

Setting $a_t = a_n \implies g^2 t = gu \implies t = \dfrac{u}{g}$.

Sol 13. Correct Option: (b)
$\vec{v}_1 = u_1\hat{i} - gt\hat{j}$, and $\vec{v}_2 = -u_2\hat{i} - gt\hat{j}$.
Perpendicular condition: $\vec{v}_1 \cdot \vec{v}_2 = 0$.

$$-u_1 u_2 + g^2 t^2 = 0 \implies t = \frac{\sqrt{u_1 u_2}}{g}$$

Sol 14. Correct Option: (c)
Horizontal separation: $x_{\text{rel}} = (u_1 + u_2)t$.
Substituting $t = \dfrac{\sqrt{u_1 u_2}}{g}$:

$$x_{\text{rel}} = \frac{(u_1 + u_2)\sqrt{u_1 u_2}}{g}$$

Sol 15. Correct Option: (d)
To hit the nth step, coordinates are $x = nb$, $y = nh$.
$nh = \frac{1}{2}gt^2 = \frac{1}{2}g\left(\frac{nb}{u}\right)^2 = \frac{gn^2b^2}{2u^2}$.
Cancelling n:

$$h = \frac{gnb^2}{2u^2} \implies n = \frac{2u^2 h}{gb^2}$$

Sol 16. Correct Option: (a)
$R = u\sqrt{\dfrac{2H}{g}}$.
Sight line angle with horizontal:

$$\tan\phi = \frac{H}{R} = \frac{\sqrt{gH}}{\sqrt{2}u} \implies \phi = \tan^{-1}\left(\frac{\sqrt{gH}}{\sqrt{2}u}\right)$$

Sol 17. Correct Option: (b)
Angle is 45° when $v_y = v_x = u \implies gt = u \implies t = \dfrac{u}{g}$.
$x = ut = \dfrac{u^2}{g}$.
$y = \dfrac{1}{2}gt^2 = \dfrac{1}{2}g\left(\frac{u}{g}\right)^2 = \dfrac{u^2}{2g}$.
Displacement:

$$r = \sqrt{x^2 + y^2} = \sqrt{\left(\frac{u^2}{g}\right)^2 + \left(\frac{u^2}{2g}\right)^2} = \frac{\sqrt{5}u^2}{2g}$$

Sol 18. Correct Option: (c)
$r(t) = \sqrt{x^2 + y^2} = \sqrt{u^2t^2 + \frac{1}{4}g^2t^4} = ut\sqrt{1 + \frac{g^2t^2}{4u^2}}$.
$\dfrac{dr}{dt} = u\left(1 + \mathcal{O}(t^2)\right) \implies \lim_{t\to 0} \dfrac{dr}{dt} = u$.

Sol 19. Correct Option: (d)
At $t = 1$ s: $v_x = 10$ m/s, $v_y = 10(1) = 10$ m/s.
Speed $v = \sqrt{10^2 + 10^2} = 10\sqrt{2}$ m/s.
Normal acceleration $a_n = g\cos(45^\circ) = \dfrac{10}{\sqrt{2}}$ m/s^2.
Radius of curvature:

$$R = \frac{v^2}{a_n} = \frac{(10\sqrt{2})^2}{10/\sqrt{2}} = \frac{200}{10/\sqrt{2}} = 20\sqrt{2}\text{ m}$$

Sol 20. Correct Option: (a)
In standard projectile motion neglecting air resistance, gravity acts solely in the vertical direction ($a_y = g$, $a_x = 0$).